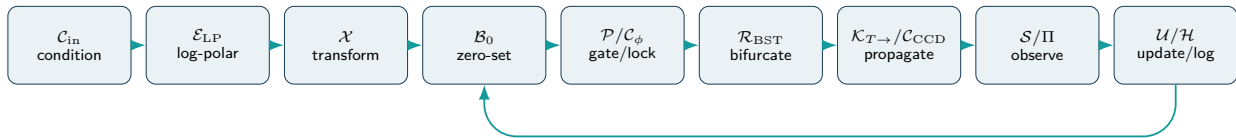


Non-Euclidean Holographic Data Fields

A Formal Operator Specification for a Parity-Conditioned Kinematic Foliation
and Implicit Holographic Co-Processor



Core thesis

A valid memory state is represented as a point on a cell-local implicit zero-level set. A causal input conditioner and log-polar address select that state. Conformal scale compensation, optional declared coordinate transforms, parity and phase-control events, bounded branch routing, forward kinematics, continuous-contact guards, analytic wavefront construction, and projection produce an observable. The observable updates the same bounded state and a tamper-evident timeline. The closed update is the self-referential engine.

Tom Klootwijk

Formal specification, Version 0.2

31 August 2026

Document status: conceptual research specification. It formalizes the supplied source materials and adds mathematically explicit commute-derived extensions. It does not claim peer review, patent status, hardware validation, police endorsement, physical realizability, or application safety.

Author declaration and dedication

Author	Tom Klootwijk
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Status of identity data	Included at the author's explicit request; not independently verified and not a cryptographic credential.

Dedication

Dedicated to police and emergency-service personnel who react, remain attentive to the public, and accept responsibility for public safety. The dedication expresses the author's intent. It does not imply adoption, validation, affiliation, or endorsement by any police organization.

The author-supplied identifier is bibliographic metadata only. A conforming implementation **MUST** not use a raw name, date of birth, or civil identifier as a cryptographic key, random seed, access token, or integrity secret. The provenance profile in this specification instead uses ordinary cryptographic key management, hash chaining, optional digital signatures, and privacy-preserving key identifiers.

Evidence boundary

The commute transcript contains time-stamped observations, personal impressions, location references, and generated technical claims. Version 0.2 extracts only the engineering ideas that can be stated as operators, invariants, hypotheses, or application profiles. The transcript itself is not experimental validation, a chain-of-custody record, or proof that the proposed engine works.

Abstract

Version 0.1 established three linked artifacts. NHDF is a representation in which each valid lookup-table cell belongs to a non-degenerate local zero-level set. PCKF is the parity-conditioned dynamical law joining log-polar jitter addressing, forward-time kinematics, phase-accelerated branching, local boundary restoration, analytic sweeps, and feedback. IHCP is the bounded runtime that executes the declared operator order on conventional, optical, quantum, or hybrid hardware.

Version 0.2 retains that core and formalizes the technically useful material developed in the 40-page commute transcript from IJburg to Hoofddorp. The principal additions are: a scale-correct log-polar Jacobian and acceleration law; finite treatment of the apex singularity; precise semantics for Möbius inversion, log-spiral shear, observer-frame compensation, and optional hyperbolic gyrotranslation; a phase-lock and orientation-transition controller; budgeted foveated attention and interrupt/recovery modes; adaptive causal step control; conservative continuous collision detection and physically defined collision response; photometric-shock and acoustic-wavefront adapters; an anisotropic region-of-interest operator; a tamper-evident provenance chain; and a public-safety application profile with mandatory human authority, privacy, auditability, and fail-safe constraints.

The upgraded document also corrects several overstatements in the transcript. Bitwise NOT is not automatically Möbius inversion; one-bit parity is not error correction; a log-polar grid does not make acceleration and deceleration identical; an SDF lookup does not make continuous collision detection free or exact without assumptions; coordinate inversion cannot prevent sensor blooming; a photographed sign is not cryptographic proof; and non-orientable topology does not create resilience by itself. These corrections preserve the source's operator vocabulary while making the model implementable and falsifiable.

Revision 0.2 in one sentence

The commute extension turns a set of evocative register-level metaphors into a typed conformal state machine with explicit coordinate metrics, bounded contact mechanics, multimodal adapters, provenance, and responsible public-safety constraints.

Revision summary

Area	Source insight	Version 0.2 formalization
Log-polar motion	A bit shift needs radial magnitude and cross-coupling.	General-base log-polar Jacobian, metric, velocity, acceleration, quantization error, and apex regularization.
Coordinate operators	Möbius inversion, gyrovector shift, ray twisting, frame drift.	Typed optional transforms with domains, equations, inverse maps, and code-equivalence requirements.
Time and phase	Temporal guard and phase lock.	Monotonic generation tags, adaptive step bound, discrete phase-locked loop, debounce, and orientation-chart transition.
Attention	Foveal concentration, interrupt, afterimage/coherence recovery.	Budgeted attention allocator, nonzero peripheral floor, snapshot ring, and causal reintegration filter.
Contact	SDF-based continuous collision detection and non-elastic response.	Conservative advancement under Lipschitz/rate bounds, contact bracketing, boundary projection, restitution, friction, sliding, and bounded fragmentation.
Sensor shock	Tunnel-to-daylight transition and anisotropic slit.	Log-luminance exposure adapter, saturation diagnostics, quality flags, and explicit ROI mask with loss accounting.
Acoustics	Police siren as Doppler and interrupt signal.	Microphone-array phase adapter, propagation-delay field, Doppler relation, and uncertainty-aware event profile.
Provenance	Blackboard/chalk/photo metadata and identity lock.	Canonical event records, hash chaining, optional signatures and trusted timestamps; raw identity is excluded from key material.
Public safety	Smarter autonomous police agents and “digital jiu-jitsu.”	Minimum-intervention constrained control, human-in-command, no autonomous use of force, audit, privacy, fairness, cybersecurity, and safe mode.

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Paradigm identity, scope, and source boundary

1.1 Three names for three layers

The upgraded specification retains the layered identity introduced in Version 0.1.

Table 1.1: Layered identity of the paradigm.

Name	Role	Formal meaning
NHDF	Representation	A bounded lookup-table and active-cell model in which each valid state is embedded in a cell-local implicit zero-level set and may cross declared non-Euclidean chart transitions.
PCKF	Dynamics	A causal hybrid update law joining log-polar addressing, conformal metric compensation, parity and phase events, bounded branching, kinematics, boundary/contact control, and feedback.
IHCP	Runtime	A software or physical architecture that executes the typed operator chain, enforces resource and safety bounds, emits telemetry, and records the next state in a tamper-evident timeline.

1.2 Source materials

This document is grounded in three supplied artifacts.

- [S0] The original 30-page prompt-and-response transcript that introduced the chained vocabulary: parity, lower-case phi, log-polar jitter, kinematic calculus, an analytic cone, projections, distributed apex, BST/L-system branching, and a Klein-bottle/SDF-zero substrate.
- [S1] The 26-page *NHDF Formal Specification, Version 0.1*, which converted that vocabulary into a non-degenerate local-zero-set state machine and separated core claims from speculative extensions.
- [S2] The 40-page commute transcript, time-stamped approximately 07:50–09:03, which proposed missing scale encoding, extra transforms, continuous contact, attention, photometric and acoustic events, provenance, and public-safety applications.

The commute transcript is treated as a design notebook: useful ideas are admitted only after they receive a defined domain, codomain, equation or algorithm, resource bound, failure mode, and validation requirement.

1.3 Distinctive contribution

The paradigm is not any single ingredient. Signed distance fields, lookup tables, parity, L-systems, phase tracking, conservative advancement, hash chains, and log-polar coordinates are established techniques. The proposed contribution is their ordered closure under a common cell state and timeline:

1. input and causal history select a log-polar address;
2. the coordinate metric converts address-space motion into physical-space motion;
3. the selected candidate is mapped to a valid cell-local zero set;
4. parity, topology, phase, and resource conditions produce a bounded control event;

5. a branch router and forward integrator update distributed apices;
6. a contact guard prevents invalid time advancement under declared assumptions;
7. geometric or wavefront operators create an observable;
8. the observable and residual update the same state; and
9. the transition is appended to a tamper-evident causal record.

1.4 Normative core, profiles, and hypotheses

Table 1.2: Status classes used in Version 0.2.

Class	Included	Excluded until validated
Normative core	Typed state, local zero-set semantics, declared operator order, log-polar metric, parity meaning, causal time, bounded resources, feedback, failure exposure.	Any universal performance or robustness claim.
Conformance profile	Optional coordinate transforms, attention, collision, photometric, acoustic, provenance, public-safety, GPU, optical, or quantum adapter.	Equating profile inclusion with superiority.
Engineering hypothesis	Quantization advantage, recovery latency, contact efficiency, foveated savings, focus quality, tamper evidence, or operator-equivalent physical transfer.	Claims of proof before measurement against a baseline.
Prohibited inference	None.	Storm immunity, zero-cost computation, automatic error correction, automatic surveillance accuracy, autonomous coercive authority, or cryptographic proof from unsigned observations.

Evidence boundary

A profile name is not a performance result. An implementation may describe itself as conforming to a profile only when it satisfies the profile's semantics, bounds, telemetry, and failure conditions. It may claim an advantage only after a controlled comparison.

Semantic commitments and notation

2.1 Disambiguation of phi, angle, scale, and time

The sources overload lower-case phi. Version 0.2 separates the meanings.

Table 2.1: Core notation.

Symbol	Term	Definition
φ_g	Golden-ratio scalar	$\varphi_g = (1+\sqrt{5})/2$; an explicitly declared branch scale or priority parameter.
$\phi_i[n]$	Signal or cell phase	Wrapped or unwrapped phase associated with cell i at generation n .
θ	Polar direction	Geometric angle in a selected two-dimensional plane.
ρ_J	Log jitter magnitude	$\rho_J = \log(1 + \gamma J)$; an address coordinate derived from residual magnitude.
w	Log spatial radius	$w = \log_b(r/r_0)$, with declared base $b > 1$, reference radius r_0 , and finite clamp.
$\Delta^2\phi$	Phase acceleration	Second finite difference of phase, after declared unwrapping and filtering.
$T \rightarrow$	Forward timeline	A monotonically increasing logical generation with positive elapsed time and no future-state read.
$F_i(z, t) = 0$	Local validity condition	Cell i 's candidate state lies on a nontrivial local zero-level set.
K	Klein-bottle topology	An intrinsic quotient/chart-transition rule, not a claim that one global Euclidean SDF exists.

2.2 Required non-degenerate reading of SDF zero

If one global scalar field is identically zero, then its gradient vanishes wherever it is differentiable. It supplies no distance, normal, interior/exterior label, or restoring direction. Version 0.2 therefore preserves the distributed reading:

$$F_i : Z_i \times \mathbb{T} \rightarrow \mathbb{R}, \quad (2.1)$$

$$M_i(t) := \{z \in Z_i : F_i(z, t) = 0\}, \quad (2.2)$$

$$z_i(t) \in M_i(t) \quad \text{for each active cell } i. \quad (2.3)$$

A field called an SDF **MUST** satisfy its declared distance approximation and gradient conditions. Otherwise it is an implicit constraint or residual field and is named as such.

2.3 Discrete time as reference semantics

The normative timeline is

$$n \in \mathbb{N}, \quad T_{n+1} > T_n, \quad \Delta t_n = T_{n+1} - T_n > 0. \quad (2.4)$$

Every state-bearing buffer slot carries a logical generation. Physical ring-buffer wrap does not reverse time. A transition computing generation $n + 1$ may read only generations at most n .

2.4 Cell state

A Version 0.2 cell is the typed record

$$c_i = (q_i, u_i, a_i, v_i, \alpha_i, \phi_i, \Delta\phi_i, \Delta^2\phi_i, \kappa_i, p_i, o_i, \nu_i, b_i, \chi_i), \quad (2.5)$$

where q_i is the packed payload; u_i the log-polar/time address; a_i, v_i, α_i apex position, velocity, and acceleration; κ_i the ordering key; p_i a parity event; $o_i \in \{+1, -1\}$ a local orientation; ν_i links, chart, epoch, and allocation metadata; b_i a resource/attention weight; and χ_i a profile and quality-flag word.

2.5 The apex is a field, not a singular global point

The active apex set is

$$\mathcal{A}_n = \{a_i[n] : i \in I_n\}. \quad (2.6)$$

A weighted centroid may be observed, but no single apex is the sole anchor. The log-polar origin requires a finite implementation rule, developed in [chapter 5](#).

Operator algebra and canonical closure

3.1 Typed operators

Table 3.1: Normative and profile operator signatures.

Operator	Signature	Role
$\mathcal{C}_{\text{in}}$	$X \times H \rightarrow X'$	Sensor calibration, photometric normalization, observer-frame compensation, and quality flags; profile-dependent.
$\mathcal{E}_{\text{LP}}$	$X' \times H \rightarrow \Omega_{\rho\theta T}$	Causal residual, log magnitude, polar direction, forward-time address, and saturation status.
$\mathcal{A}_B$	$\Omega \times L \rightarrow \mathcal{I}_B$	Budgeted attention/resource selection with a nonzero peripheral monitoring floor.
$\mathcal{X}_{\xi}$	$Z \rightarrow Z$	Declared optional coordinate transform selected from a typed profile: identity, Möbius inversion, log-spiral shear, or hyperbolic transport.
$\mathcal{B}_0$	$Z_i \rightarrow M_i$	Project or embed a candidate on a cell-local zero set within tolerance.
$\mathcal{P}$	$\mathbb{B}^w \times \{\pm 1\} \rightarrow \mathbb{B}^k$	Data parity, topology orientation parity, or diagnostic pair; $k = 1$ preserves source constraint.
$\mathcal{C}_{\phi}$	$\Phi \times \mathbb{B}^k \rightarrow C$	Phase lock, debounce, hysteresis, orientation-chart transition, and safe-mode request.
$\mathcal{R}_{\text{BST}}$	$C \times B \times \mathbb{R} \rightarrow \mathcal{T}$	BST ordering and L-system bifurcation at declared branch nodes under capacity bounds.
$\mathcal{K}_{T \rightarrow}^J$	$\mathcal{T} \times \Delta t \rightarrow \mathcal{T}$	Forward kinematics using the declared log-polar Jacobian, physical-space acceleration, damping, and generation semantics.
$\mathcal{C}_{\text{CCD}}$	$\mathcal{T} \times F \rightarrow \mathcal{T}$	Conservative step bound, contact localization, boundary correction, and declared collision response.
$\mathcal{S}$	$\mathcal{T} \rightarrow G$	Analytic cone, implicit geometry, acoustic delay, or other declared wavefront/field construction.
Π	$G \rightarrow Y$	Image, projection, beam, field slice, feature, or control observable.
$\mathcal{U}$	$L \times Y \times R \rightarrow L$	Feedback, allocation, residual, quality, and generation update.
$\mathcal{H}$	$E \times h \rightarrow h'$	Canonical event digest and append-only provenance-chain update.

3.2 Expanded canonical chain

Let L_n be the complete bounded state and let h_n be the provenance-chain head. The Version 0.2 output is

$$y_n = \Pi \circ \mathcal{S} \circ \mathcal{C}_{\text{CCD}} \circ \mathcal{K}_{T \rightarrow}^J \circ \mathcal{R}_{\text{BST}}^{\varphi_g, \Delta^2 \phi_n} \circ \mathcal{C}_{\phi} \circ \mathcal{P} \circ \mathcal{B}_0 \quad (3.1)$$

$$\circ \mathcal{X}_{\xi_n} \circ \mathcal{A}_B \circ \mathcal{E}_{\text{LP}} \circ \mathcal{C}_{\text{in}}(x_n; L_n). \quad (3.2)$$

The closed state and provenance update is

$$L_{n+1} = \mathcal{U}(L_n, y_n, r_n), \quad (3.3)$$

$$h_{n+1} = \mathcal{H}(h_n, E_{n+1}), \quad (3.4)$$

$$(L_{n+1}, h_{n+1}) = \mathfrak{N}_{\Delta t}(x_n; L_n, h_n). \quad (3.5)$$

The coordinate transform $\mathcal{X}_{\xi_n}$ is the identity unless a conformance profile declares another transform, its trigger, and its inverse or fallback.

3.3 Normative precedence

Unless an implementation proves semantic equivalence of a fused stage, the order is:

1. acquire, calibrate, tag quality, and stabilize the observer frame;
2. derive causal residuals and log-polar/time addresses;
3. select active cells under a finite attention and work budget;
4. apply the declared coordinate-transform block;
5. project candidates to local zero sets;
6. compute data/topology parity and phase-control state;
7. perform bounded branch routing;
8. integrate forward kinematics with the log-polar metric;
9. limit advancement and resolve contact;
10. construct wavefront/implicit geometry and project an observable;
11. feed the residual and output into the next bounded state; and
12. append a canonical event digest to the causal provenance chain.

Normative requirement

Changing operator order is a semantic change, not an optimization detail. A reordered implementation **MUST** publish the new composition and show, with reference vectors, when it is equivalent and when it is not.

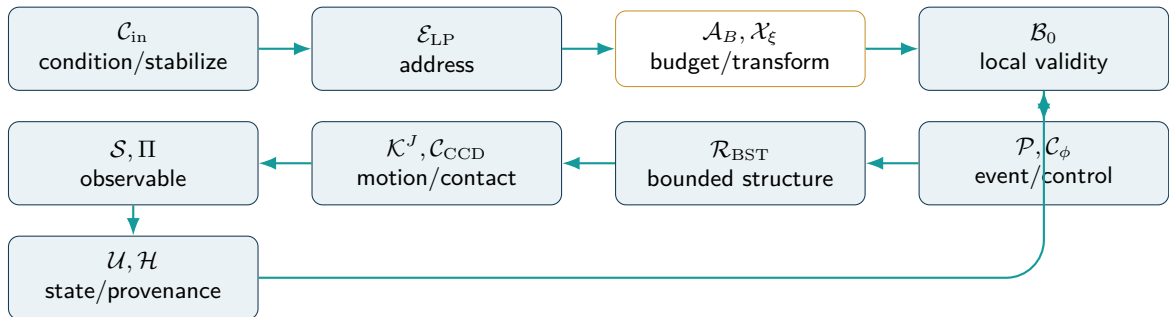


Figure 3.1: Version 0.2 operator chain. Gold stages are optional profiles; their identity behavior preserves the core chain.

Local zero-set substrate and topology

4.1 Boundary projection

For a differentiable cell-local field with nonzero gradient near its zero set, one regularized projection step is

$$\mathcal{B}_0(z) = z - \lambda_B \frac{F_i(z, T_n)}{\|\nabla F_i(z, T_n)\|_2^2 + \varepsilon_B} \nabla F_i(z, T_n), \quad (4.1)$$

with $0 < \lambda_B < 2$ and $\varepsilon_B > 0$. The residual is

$$r_{B,i} = |F_i(z_i, T_n)|. \quad (4.2)$$

An implementation uses a finite tolerance $r_{B,i} \leq \tau_B$ and records singular or ill-conditioned gradients.

4.2 Klein-bottle chart rule

The intrinsic Klein bottle is represented by the quotient

$$(0, v) \sim (1, 1 - v), \quad (u, 0) \sim (u, 1). \quad (4.3)$$

Crossing the first seam reverses a chart coordinate and therefore flips a local orientation sign. This supplies a concrete topological event; it does not supply error correction or a globally signed Euclidean distance.

4.3 Data parity and topology parity

For payload $q_i \in \mathbb{B}^w$,

$$p_i^{\text{data}} = \bigoplus_{b=0}^{w-1} \text{bit}_b(q_i). \quad (4.4)$$

For a chart path Γ_i with transition orientations $\sigma_e \in \{+1, -1\}$,

$$p_i^{\text{topo}} = \frac{1 - \prod_{e \in \Gamma_i} \sigma_e}{2}. \quad (4.5)$$

The one-bit source-compatible gate may be $p_i = p_i^{\text{data}} \oplus p_i^{\text{topo}}$. A diagnostic profile stores both bits separately so that corruption and orientation events are not conflated.

Evidence boundary

XOR parity detects an odd number of inversions between checks, misses every even number, cannot locate the damaged bit, and cannot repair it. Any integrity claim requires stronger coding, redundancy, recomputation, or an application residual.

Conformal log-polar kinematics

5.1 Why the scale metric is required

The commute transcript correctly identifies a missing element in a bit-shift interpretation of log-polar motion: one address step does not correspond to one constant physical displacement. Let

$$w = \log_b(r/r_0), \quad r = r_0 b^w, \quad b > 1, \quad (5.1)$$

and let θ be the geometric polar angle. In a two-dimensional plane,

$$p(w, \theta) = p_0 + r_0 b^w \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}. \quad (5.2)$$

The Jacobian is

$$J_{LP}(w, \theta) = r \begin{bmatrix} (\ln b) \cos \theta & -\sin \theta \\ (\ln b) \sin \theta & \cos \theta \end{bmatrix}. \quad (5.3)$$

For natural logarithms, $\ln b = 1$ and $J_{LP} = rR(\theta)$. For base two, the radial derivative carries the factor $\ln 2$ even if an integer implementation uses shifts to synthesize 2^w .

The induced metric is

$$G_{LP} = J_{LP}^T J_{LP} = r^2 \text{diag}((\ln b)^2, 1), \quad (5.4)$$

and the area scale is

$$|\det J_{LP}| = (\ln b) r^2. \quad (5.5)$$

Consequently, an equal increment in w or θ has a radius-dependent physical meaning. This scale factor is the rigorous version of the transcript's “radial magnitude” or “Jacobian” correction.

5.2 Velocity and acceleration

Let $e_r = (\cos \theta, \sin \theta)^T$ and $e_\theta = (-\sin \theta, \cos \theta)^T$. The physical velocity is

$$\dot{p} = r((\ln b)\dot{w} e_r + \dot{\theta} e_\theta). \quad (5.6)$$

The physical acceleration is

$$\ddot{p} = r((\ln b)\ddot{w} + (\ln b)^2 \dot{w}^2 - \dot{\theta}^2) e_r + (\ddot{\theta} + 2(\ln b)\dot{w}\dot{\theta}) e_\theta. \quad (5.7)$$

The $-\dot{\theta}^2$ term is centripetal and the $2(\ln b)\dot{w}\dot{\theta}$ term is the radial-angular cross coupling that an independent barrel shift omits. A conforming kinematic profile **MUST** either evaluate these terms, use a numerically equivalent Cartesian update, or declare a bounded approximation and its error.

5.3 Discrete update

For a cell with log-polar state $\zeta_i = (w_i, \theta_i)$ and physical state (a_i, v_i) , two valid implementation strategies are:

1. integrate a_i, v_i in Cartesian space and encode the result back into (w_i, θ_i) ; or
2. integrate ζ_i using the metric, Christoffel terms, and a declared manifold integrator.

The reference implementation uses Cartesian semi-implicit integration:

$$v_i[n+1] = v_i[n] + \Delta t_n \alpha_i[n], \quad (5.8)$$

$$a_i[n+1] = a_i[n] + \Delta t_n v_i[n+1], \quad (5.9)$$

$$(w_i[n+1], \theta_i[n+1]) = \text{encode}_{\text{LP}}(a_i[n+1] - p_0). \quad (5.10)$$

This avoids pretending that an address shift alone is a complete physical integrator.

5.4 Quantization and radial error

Let $\hat{w} = Q_w(w)$ with maximum log-coordinate error $|\hat{w} - w| \leq \Delta w/2$. The relative radial error obeys

$$\left| \frac{\hat{r} - r}{r} \right| \leq b^{\Delta w/2} - 1. \quad (5.11)$$

Angular quantization with $|\hat{\theta} - \theta| \leq \Delta\theta/2$ contributes approximately $r\Delta\theta/2$ tangential error at small angles. Version 0.2 therefore requires calibration of both radial and angular quantizers rather than assuming that a fixed bit width is uniformly accurate.

5.5 Apex regularization

The polar apex $r = 0$ corresponds to $w = -\infty$ and cannot be represented in finite registers. A conforming implementation **MUST** declare one of the following:

- a finite clamp $r \geq r_{\min} > 0$, with $w_{\min} = \log_b(r_{\min}/r_0)$;
- an apex sentinel with a separate local Cartesian micro-chart; or
- a multi-chart atlas that excludes the singular coordinate.

Saturation at $w_{\min}$ is logged. No claim of infinite foveal resolution is permitted on finite hardware.

Correction to a commute claim

Moving inward or outward on a log-polar grid changes spatial resolution and quantization. It does not, by itself, define physical acceleration or deceleration. Acceleration remains the derivative of physical velocity and must obey the Jacobian-transformed law above.

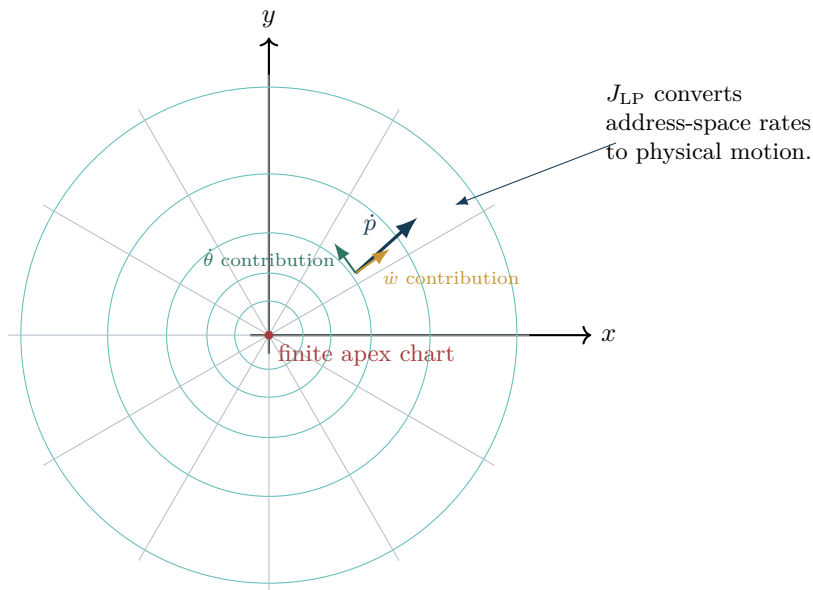


Figure 5.1: Log-polar cells expand with radius. The Jacobian and cross terms are part of the dynamics, not optional visual decoration.

6

Declared coordinate-transform profiles

6.1 Transform block semantics

The commute transcript introduces several named operators as if each were a bitwise instruction. Version 0.2 admits them only as typed coordinate transforms. The transform block is

$$\mathcal{X}_\xi = \mathcal{G}_c \circ \Gamma_\kappa \circ \mathcal{M}_{c,R} \quad (6.1)$$

when all three profiles are enabled; disabled stages are the identity. This order is canonical: inversion first establishes scale/orientation, log-spiral shear then couples radial and angular coordinates, and geodesic transport finally translates in the declared metric. An implementation may choose a different order only by publishing it as a distinct profile.

6.2 Möbius inversion profile

Represent a planar coordinate by $z = x + iy$. The source form $z \mapsto 1/z$ is generalized to

$$\mathcal{M}_{c,R}(z) = c + \frac{R^2}{z - c}, \quad z \neq c. \quad (6.2)$$

For $c = 0$, $R = 1$, and $z = re^{i\theta}$,

$$r' = r^{-1}, \quad \theta' = -\theta, \quad (6.3)$$

so in log-polar coordinates

$$w' = -w, \quad \theta' = -\theta \pmod{2\pi}. \quad (6.4)$$

With a general inversion radius, $w' = 2 \log_b(R/r_0) - w$. The map is involutive away from its pole.

A local field transforms by pullback:

$$F_i^{\mathcal{M}}(z, t) = F_i(\mathcal{M}_{c,R}^{-1}(z), t). \quad (6.5)$$

Thus the zero set is transformed without claiming that the field values remain exact signed distances.

Register-equivalence requirement

A bitwise NOT is not generally Möbius inversion. For a fixed-point register $q = \text{enc}(w)$, an implementation **MUST** show that its register operation satisfies $\text{dec}(M_q(q)) = -\text{dec}(q) + c$ within tolerance. In two's-complement arithmetic, negation is normally $\sim q + 1$, not merely $\sim q$.

6.3 Log-spiral shear profile

The commute transcript's “geodesic ray-twister” is made concrete as an invertible shear in log-polar coordinates:

$$\Gamma_{\kappa,\delta}(w, \theta) = (w + \delta_w, \text{wrap}(\theta + \kappa w + \delta_\theta)). \quad (6.6)$$

Its Jacobian in (w, θ) is

$$D\Gamma = \begin{bmatrix} 1 & 0 \\ \kappa & 1 \end{bmatrix}, \quad \det D\Gamma = 1. \quad (6.7)$$

Lines of constant transformed phase correspond to logarithmic spirals in physical space. A dynamic twist may be driven by a bounded second phase difference,

$$\kappa_i[n] = \text{clip}(\kappa_0 + k_\kappa \Delta^2 \phi_i[n], -\kappa_{\max}, \kappa_{\max}). \quad (6.8)$$

The shear couples scale and direction but does not replace physical frame compensation or contact detection.

6.4 Optional hyperbolic gyrotranslation

If, and only if, the state is explicitly mapped into a Poincaré ball $\mathbb{D}_c^d = \{x : c\|x\|^2 < 1\}$ of curvature $-c$, $c > 0$, a gyrovector translation may use Möbius addition

$$u \oplus_c v = \frac{(1 + 2c\langle u, v \rangle + c\|v\|^2)u + (1 - c\|u\|^2)v}{1 + 2c\langle u, v \rangle + c^2\|u\|^2\|v\|^2}. \quad (6.9)$$

The operator $\mathcal{G}_c(u; v) = u \oplus_c v$ is a research profile. It is invalid to apply the formula to arbitrary packed words without an encode/decode map, norm bound, pole handling, and reference implementation.

6.5 Composition, inverse, and quality flags

Every transform profile **MUST** publish:

- domain, codomain, parameters, and singular sets;
- forward map and, where required, inverse map;
- effect on local fields, gradients, and orientation;
- quantization and overflow behavior;
- a quality flag indicating saturation, pole proximity, or inverse failure; and
- reference vectors for identity, boundary, and extreme cases.

Input conditioning, observer frame, and photometric shock

7.1 Observer-frame compensation

The source name “Lorentz frame drift” is retained only as a historical alias. For ordinary transit, robotics, or camera motion, the formal operator is an observer-frame transform in $SE(d)$, not a relativistic Lorentz transformation.

Let $g_n = (R_n, t_n) \in SE(d)$ map sensor coordinates into a stabilized frame. A point and velocity are conditioned by

$$p_n^{\text{stab}} = R_n p_n^{\text{sensor}} + t_n, \quad (7.1)$$

$$v_n^{\text{rel}} = v_n^{\text{object}} - v_n^{\text{frame}}. \quad (7.2)$$

In a rotating three-dimensional frame, the implementation includes the standard angular-velocity terms or performs the complete homogeneous transform before differencing. The estimated frame state and its uncertainty are part of the input quality record.

7.2 Stationary calibration window

A low-motion interval can calibrate sensor bias, clock offset, background phase, and exposure. Define a stationary gate

$$q_{\text{stat}}[n] = \mathbf{1} [\|\hat{v}_{\text{frame}}[n]\| \leq \tau_v \wedge \|\hat{\alpha}_{\text{frame}}[n]\| \leq \tau_a]. \quad (7.3)$$

A calibration update is accepted only after $q_{\text{stat}} = 1$ for a declared dwell time and after residual checks pass. Stopping at a platform or workstation can therefore be a calibration opportunity, but the location itself is not a calibration certificate.

7.3 Photometric shock adapter

A sudden transition from dark to bright conditions is a sensor and photometric problem. Let $Y_n(x)$ be luminance and

$$\ell_n(x) = \log(\varepsilon_Y + Y_n(x)). \quad (7.4)$$

Define saturation fraction and shock magnitude

$$s_n = \frac{1}{|\Omega|} \sum_{x \in \Omega} \mathbf{1}[Y_n(x) \geq Y_{\text{max}}], \quad (7.5)$$

$$q_n = |\text{median}(\ell_n) - \text{median}(\ell_{n-1})|. \quad (7.6)$$

When $s_n > \tau_s$ or $q_n > \tau_q$, the input conditioner may freeze fragile state updates, lower exposure/gain, select an HDR path, and emit a quality flag. Spatial Möbius inversion may reallocate coordinates, but it cannot remove charge overflow, blooming, clipping, or missing photons in a physical sensor.

Evidence boundary

Anti-blooming, dynamic range, rolling shutter, and exposure are properties of the sensor and acquisition pipeline. A coordinate transform or NOT gate cannot guarantee their correction. Any camera claim requires measured raw-sensor and reconstructed-image results.

7.4 Anisotropic region-of-interest operator

The commute transcript’s narrow “vision slit” becomes an explicit mask

$$m_n(x, y) \in \{0, 1\}, \quad X_n^{\text{ROI}}(x, y) = m_n(x, y)X_n(x, y). \quad (7.7)$$

For a horizontal strip, $m_n = 1$ when $|y - y_f| \leq h/2$. The operator reduces work in proportion to the retained samples, but it also removes information. A conforming profile records mask geometry, dropped coverage, and the safety conditions under which the mask may be used.

Phase control, parity conditioning, and orientation

8.1 Second-order phase variation

After phase unwrapping or a declared wrapped-difference method,

$$\Delta^2\phi_i[n] = \text{wrap}(\phi_i[n] - 2\phi_i[n-1] + \phi_i[n-2]). \quad (8.1)$$

Because second differences amplify noise, the input phase **SHOULD** be filtered and its sample rate justified. Branch angle and scale may be modulated by

$$\beta_i[n] = \beta_0 + k_\phi \text{clip}(\Delta^2\phi_i[n], -\delta_{\max}, \delta_{\max}), \quad (8.2)$$

$$s_i[n] = \varphi_g^{-\eta} \exp(-k_s |\Delta^2\phi_i[n]|), \quad \eta > 0. \quad (8.3)$$

The default $s_i < 1$ supports bounded recursive length.

8.2 Discrete phase-locked loop

Let ϕ_i^{meas} be measured phase and $\hat{\phi}_i$ the predicted phase. Define

$$e_{\phi,i}[n] = \text{wrap}(\phi_i^{\text{meas}}[n] - \hat{\phi}_i[n]). \quad (8.4)$$

A reference second-order digital lock is

$$\hat{\omega}_i[n+1] = \text{clip}(\hat{\omega}_i[n] + k_I e_{\phi,i}[n], -\omega_{\max}, \omega_{\max}), \quad (8.5)$$

$$\hat{\phi}_i[n+1] = \text{wrap}(\hat{\phi}_i[n] + \Delta t_n \hat{\omega}_i[n+1] + k_P e_{\phi,i}[n]). \quad (8.6)$$

The lock state becomes true only when $|e_{\phi,i}| \leq \tau_\phi$ for a declared dwell count. A parity event may request reacquisition; it is not itself a phase estimator.

8.3 Debounce, hysteresis, and event rate

The raw parity bit may be filtered by a causal vote

$$\bar{p}_i[n] = \mathbf{1} \left[\sum_{r=0}^{h-1} p_i[n-r] \geq m \right], \quad 1 \leq m \leq h. \quad (8.7)$$

The toggle rate is

$$f_{p,i}[n] = \frac{1}{h\Delta t} \sum_{r=1}^h |p_i[n-r+1] - p_i[n-r]|. \quad (8.8)$$

Excess rate, saturation, unlock, or resource pressure enters safe mode rather than driving uncontrolled branch oscillation.

8.4 Orientation transition and the y-axis example

A coordinate sign changes only when a declared chart transition or application transform requires it. If $o_i \in \{+1, -1\}$ is the chart orientation, a two-dimensional projection may use

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & o_i \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}. \quad (8.9)$$

A parity bit can report the odd/even count of orientation reversals, but it does not arbitrarily flip a sign merely because a clock digit changes.

Interpretation of the 07:50–07:51 observation

The observed time-digit transition is useful as a human-readable example of a discrete edge and generation change. In the formal engine it motivates monotonic generation tags and event logging, not a physical proof that the right-hand bit controls a spatial axis.

Budgeted attention, interrupt, and recovery

9.1 Foveation as resource allocation

The commute transcript’s attention language is formalized as an engineering allocation policy, not a diagnosis or model of a person. Let $d_{LP}(u_i, u_f)$ be a declared distance in log-polar address space from cell i to a task-selected focus u_f . Define a saliency score

$$s_i = \lambda_q q_i^{\text{task}} + \lambda_r \exp\left(-\frac{d_{LP}(u_i, u_f)^2}{2\sigma_f^2}\right) + \lambda_e q_i^{\text{event}}, \quad (9.1)$$

where q_i^{task} , q_i^{event} , and all weights are application-defined and logged.

The allocator chooses activity levels $b_i \in [b_{\min}, 1]$ under a finite budget:

$$\max_{b_i} \sum_i s_i b_i \quad \text{subject to} \quad \sum_i c_i b_i \leq B_n, \quad b_i \geq b_{\min} > 0 \text{ for safety-relevant peripheral cells.} \quad (9.2)$$

The nonzero floor prevents total blindness outside the focus. “Perfect attention at zero effort” is therefore not a normative claim; finite sensing and computation always have cost.

9.2 Interrupt state

A high-priority event may enter a focused mode when

$$q_i^{\text{event}} > \tau_e \quad \text{or} \quad r_i^{\text{safety}} < \tau_{\text{near}} \quad \text{or} \quad q_n^{\text{shock}} > \tau_q. \quad (9.3)$$

The interrupt handler performs four bounded actions:

1. store the current generation and active-cell digest in a ring-buffer snapshot;
2. reserve a fixed work budget for the event region;
3. reduce, but do not eliminate, background update rates; and
4. set an expiration time and revalidation condition.

No unbounded recursive branch or hidden allocation is permitted during an interrupt.

9.3 Causal coherence recovery

Let z_i^{cache} be the last valid background state and z_i^{new} the reacquired state. In Euclidean components a simple recovery filter is

$$z_i^{\text{rec}}[n+1] = (1 - \alpha_i) z_i^{\text{cache}} + \alpha_i z_i^{\text{new}}, \quad 0 < \alpha_i \leq 1. \quad (9.4)$$

For chart or manifold coordinates, interpolation uses a declared geodesic or local chart and checks seam orientation. Recovery is complete only after boundary residual, phase lock, and generation consistency pass.

9.4 Attention telemetry

A conforming run records the focus address, active budget, peripheral floor, interrupt cause, dwell time, dropped samples, recovery residual, and any safety-region coverage loss. This turns the transcript’s “afterimage” or “distraction latch” into a reproducible state transition rather than an unverifiable mental-state claim.

10

Golden-ratio BST/L-system router

10.1 Grammar and ordering

Let $B(d, \ell, \psi, \kappa)$ denote a branch node at depth d , length ℓ , direction ψ , and ordering key κ . A gated bifurcation is

$$B(d, \ell, \psi, \kappa) \xrightarrow{\bar{p}=1} \begin{cases} B(d+1, s\ell, \psi + \beta, \kappa_L), \\ B(d+1, s\ell, \psi - \beta, \kappa_R), \end{cases} \quad (10.1)$$

subject to

$$\kappa_L < \kappa_P < \kappa_R. \quad (10.2)$$

The geometric L-system rule and BST ordering are distinct constraints applied to the same node records.

10.2 Bounded allocation

The active node count obeys

$$0 \leq N_n \leq N_{\max}. \quad (10.3)$$

A deterministic overflow policy must select among rejecting low-priority proposals, recycling expired nodes, merging nearby leaves, or reducing branch rate. Branch allocation is a separate scan/queue stage; it may not be hidden in a function that can silently overrun memory.

10.3 Golden ratio as a hypothesis

The golden ratio is preserved because it is part of the source definition, but it is not privileged by proof. A conformance profile declares exactly how φ_g affects length, angle, priority, or quantization. Validation must compare it with fixed, optimized, random, and application-specific alternatives at equal node and compute budgets.

11

Adaptive forward timeline and workload guard

11.1 Directional adherence

Every update has a strictly increasing generation g_n . A physical buffer slot stores (g, z) , and a read for generation g_n rejects any slot tagged $g > g_n$. This implements the source's one-dimensional forward timeline without claiming zero latency or forbidding bounded ring-buffer wrap.

11.2 Resolution-aware time step

For a log-polar cell at radius r_i , a conservative local physical resolution is

$$\ell_i = r_i \min((\ln b)\Delta w, \Delta\theta). \quad (11.1)$$

Let V_i and A_i bound relative speed and acceleration over the next step. A resolution-aware step satisfies

$$V_i\Delta t + \frac{1}{2}A_i\Delta t^2 \leq \eta_\ell \ell_i, \quad 0 < \eta_\ell < 1. \quad (11.2)$$

Thus

$$\Delta t_i \leq \begin{cases} \frac{\sqrt{V_i^2 + 2A_i\eta_\ell \ell_i} - V_i}{A_i}, & A_i > 0, \\ \frac{\eta_\ell \ell_i}{V_i + \varepsilon}, & A_i = 0. \end{cases} \quad (11.3)$$

The global step is the minimum of nominal, resolution, contact, sensor, and deadline bounds.

11.3 Workload governor

Let C_n estimate remaining work and D_n the deadline budget. The governor controls active cells, branch proposals, sweep samples, and output resolution so that

$$C_n \leq D_n - \Delta_{\text{reserve}}. \quad (11.4)$$

When the bound cannot be met, the engine degrades a declared noncritical output, preserves safety cells, freezes speculative branches, or enters safe mode. It does not relabel dropped computation as “time dilation.”

Table 11.1: Reference runtime modes.

Mode	Trigger	Required behavior
Calibrate	sustained low frame motion and valid sensors	Estimate bias/exposure/phase baseline; do not overwrite last trusted state until checks pass.
Transit	observer motion above threshold	Enable frame compensation and resolution-aware time step.
Focused	event/saliency threshold	Reallocate bounded work while maintaining peripheral floor.
Shock	photometric or acoustic discontinuity	Freeze fragile state, adapt frontend, retain quality flags.
Contact	predicted gap below threshold	Invoke conservative advancement and collision response.
Safe	parity rate, unlock, overflow, deadline, or residual alarm	Stop new branching, damp motion, preserve last valid generation, request external handling.
Stationary	low motion after transit	Permit calibration, compaction, and provenance checkpoint.

12

Continuous collision detection and response

12.1 Contact problem

Let $q(t)$ be a moving point or reduced configuration and let $F(q, t) > 0$ denote separation from a boundary, with $F = 0$ at contact. If F is an exact signed distance to a static obstacle, $\|\nabla F\| = 1$ almost everywhere away from singularities. In the general case, the implementation supplies a rate bound

$$B_F \geq |\partial_t F| + \|\nabla F\| V_{\text{rel}} \quad (12.1)$$

throughout the current local region.

12.2 Conservative advancement

For positive gap $g = F(q(t), t)$, a conservative step is

$$\Delta t_{\text{CCD}} = \eta_c \frac{\max(g - \tau_c, 0)}{B_F + \varepsilon_c}, \quad 0 < \eta_c < 1. \quad (12.2)$$

The engine advances by the minimum of this bound and the other timeline bounds. If $g \leq \tau_c$, if a sign change is observed, or if the rate bound is invalid, it brackets the event and applies a declared root solver such as bisection or safeguarded Newton iteration.

For known acceleration, a tighter bound may solve a quadratic closing-distance inequality. All approximations retain a safety factor and a maximum iteration count.

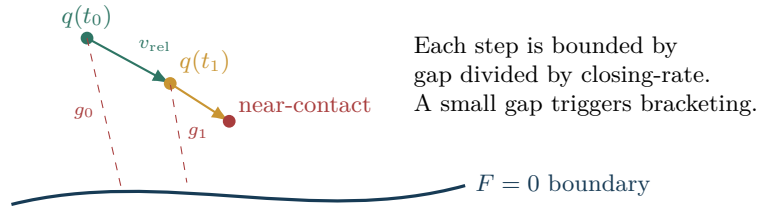


Figure 12.1: Conservative advancement. The SDF or constraint supplies a gap only under declared validity and rate assumptions.

12.3 Boundary localization and position correction

At contact, the unit normal is

$$n = \frac{\nabla F(q, t)}{\|\nabla F(q, t)\| + \varepsilon_n}. \quad (12.3)$$

Residual penetration is corrected by the local projection $\mathcal{B}_0$, with a maximum correction distance and failure flag. A singular gradient invokes a fallback normal, neighboring chart, or safe stop.

12.4 Restitution and sliding

Let v_s be boundary velocity and $v_{\text{rel}} = v - v_s$. Write

$$v_n = v_{\text{rel}} \cdot n, \quad v_t = v_{\text{rel}} - v_n n. \quad (12.4)$$

For an approaching contact $v_n < 0$, a restitution coefficient $e \in [0, 1]$ gives the normal component $-ev_n n$. A Coulomb-like tangential update is

$$v'_t = \max \left(0, 1 - \mu \frac{(1+e)(-v_n)}{\|v_t\| + \varepsilon_v} \right) v_t. \quad (12.5)$$

The outgoing velocity is

$$v' = v_s - ev_n n + v'_t. \quad (12.6)$$

Setting $e = 0$ and removing only inward normal motion gives a sliding/sticking profile. Energy loss is controlled by e and μ , not by arbitrarily discarding low-order bits.

12.5 Fragmentation profile

Fragmentation is an application event, not a universal collision response. If enabled, it **MUST** specify mass/weight distribution, momentum and energy policy, fragment count bound, branch-pool capacity, minimum fragment size, and deterministic overflow behavior. The BST router may allocate fragments, but a parity flip alone is insufficient justification.

12.6 What the CCD profile can and cannot claim

Evidence boundary

The engine does not “solve CCD in one clock” merely because it stores an SDF. Conservative advancement is reliable only when the field, rate bound, time-varying obstacle model, integration, quantization, and tolerances are valid. Thin features, discontinuous fields, moving/deforming surfaces, high acceleration, and gradient singularities require explicit tests and fallbacks.

13

Analytic sweeps, projections, and multimodal wavefronts

13.1 Cone and swept field

For apex a_i , unit axis u_i , and half-angle χ_i , an analytic cone surface is

$$C_i(x) = ((x - a_i) \cdot u_i)^2 - \cos^2(\chi_i) \|x - a_i\|^2 = 0, \quad (x - a_i) \cdot u_i \geq 0. \quad (13.1)$$

This is an implicit cone equation, not automatically an exact SDF. A swept field over a horizon H is

$$G_n(x) = \min_{i \in I_n} \min_{\tau \in [T_n - H, T_n]} \Psi(C_i(x; \tau)), \quad (13.2)$$

where Ψ converts residual to occupancy, distance estimate, intensity, phase delay, or another declared observable.

13.2 Projection vocabulary

Orthographic maps include

$$\Pi_{xy}(x, y, z) = (x, y), \quad \Pi_{yz}(x, y, z) = (y, z), \quad \Pi_{xz}(x, y, z) = (x, z). \quad (13.3)$$

A sphere projects to a circle; an aligned finite cone or pyramid has a triangular side envelope; circular cross-sections yield circles. These are calibration targets, not guaranteed output shapes for every state.

13.3 Acoustic wavefront adapter

For a source at $s(t)$ in a homogeneous medium with sound speed c , propagation delay is

$$\tau(x, t) = \frac{\|x - s(t)\|}{c}. \quad (13.4)$$

Equal-delay surfaces are spheres, or circles in a planar slice. This delay field may be used as an implicit wavefront, but acoustic pressure itself is oscillatory and is not a signed distance.

For microphone m and frequency bin f , let

$$X_m(f, n) = A_m(f, n) e^{i\phi_m(f, n)}. \quad (13.5)$$

Magnitude may be log encoded, and inter-microphone phase differences or time differences estimate direction. Second phase differences indicate frequency-rate or chirp-like behavior after unwrapping. They do not yield an exact source velocity without a source-frequency model and geometry.

13.4 Doppler relation

For collinear motion in a stationary medium, a conventional sign convention gives

$$f_{\text{obs}} = f_0 \frac{c + v_o}{c - v_s}, \quad (13.6)$$

where v_o is observer velocity toward the source and v_s is source velocity toward the observer. A siren pass can therefore serve as an acoustic calibration or event stream when f_0 , clock synchronization, multipath, and uncertainty are handled. It is not proof of the engine's correctness.

13.5 Emergency acoustic interrupt profile

A public-safety acoustic adapter may detect an emergency-signal candidate, estimate direction and confidence, and request a bounded attention interrupt. It **MUST** retain false-positive/false-negative metrics, avoid treating loudness alone as authority, and defer operational decisions to the application and human command structure.

Self-reference, stability, and failure exposure

14.1 Closed update

The feedback operator updates addressed cells, branch metadata, histories, quality flags, and output residuals:

$$L_{n+1} = \mathcal{U}(L_n, u_n, y_n, r_n, p_n, \chi_n). \quad (14.1)$$

For numeric state a relaxed update may be

$$L_{n+1}^{\text{num}} = (1 - \mu)L_n^{\text{num}} + \mu\tilde{L}_{n+1}^{\text{num}}, \quad 0 < \mu \leq 1, \quad (14.2)$$

while discrete allocation and mode fields follow declared transition rules.

14.2 Fixed point and hybrid modes

For stationary input x , a self-consistent state L^* satisfies

$$L^* = \mathfrak{N}_{\Delta t}(x; L^*). \quad (14.3)$$

In a smooth mode, local behavior can be studied through

$$J_L = \left. \frac{\partial \mathfrak{N}_{\Delta t}(x; L)}{\partial L} \right|_{L=L^*}. \quad (14.4)$$

A sufficient local condition is spectral radius $\rho(J_L) < 1$. The full engine is hybrid and nonsmooth because parity, mode, contact, and allocation events change the update map. Each mode and transition surface must be analyzed separately.

14.3 Boundary restoration

For one projection step with small residual,

$$F_i(\mathcal{B}_0(z)) = (1 - \lambda_B)F_i(z) + O(F_i(z)^2) \quad (14.5)$$

when $\nabla F_i \neq 0$ and regularization is small. Thus $0 < \lambda_B < 2$ contracts a sufficiently small local residual to first order. This is the precise form of “pulling the apex back to the boundary.”

14.4 Practical stability envelope

A run is within its declared envelope only when

$$\max_i r_{B,i} \leq \tau_B, \quad N_n \leq N_{\max}, \quad (14.6)$$

$$\max_i \|v_i\| \leq v_{\max}, \quad \max_i f_{p,i} \leq f_{\max}, \quad (14.7)$$

$$\|r_n\| \leq \tau_r, \quad C_n \leq D_n - \Delta_{\text{reserve}}, \quad (14.8)$$

$$s_n \leq \tau_s, \quad h_{\text{contact}} \geq 0 \text{ or contact mode is active.} \quad (14.9)$$

No topology, coordinate transform, or parity bit guarantees this envelope.

14.5 Failure is an output

When an assumption fails, the engine produces a quality/failure state rather than silently substituting a metaphor. Mandatory failure classes include:

- invalid or singular field gradient;
- log-radius or angle saturation;
- transform pole or inverse failure;
- phase unwrap or lock failure;
- parity-rate chatter;
- branch or memory pressure;
- contact-rate bound invalidation;
- photometric saturation or missing sensor data;
- deadline miss;
- provenance-signature or clock failure; and
- application residual outside tolerance.

Tamper-evident timeline and identity handling

15.1 From blackboard metadata to event records

The commute transcript treats signs, tags, reflections, photographs, and timestamps as “blackboard” metadata. Version 0.2 turns that idea into an append-only event model. A canonical event record is

$$E_n = (g_n, T_n, d_n^{\text{input}}, d_n^{\text{state}}, d_n^{\text{output}}, m_n, r_n, \chi_n, a_n^{\text{attach}}, k_n^{\text{actor}}), \quad (15.1)$$

where each d is a cryptographic digest, m_n is runtime mode, χ_n contains alarms/quality, a_n^{attach} is an optional attachment digest, and k_n^{actor} is a privacy-preserving key or role identifier.

15.2 Hash chain

With a collision-resistant hash H and an unambiguous canonical encoding canon,

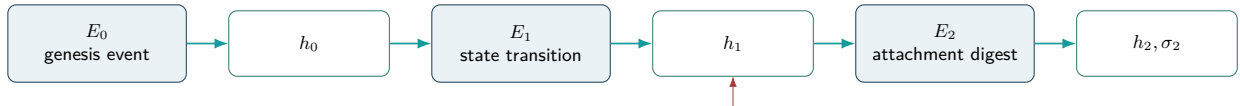
$$h_0 = H(\text{NHDF-v0.2-genesis} \parallel r_0), \quad (15.2)$$

$$h_n = H(\text{NHDF-event} \parallel h_{n-1} \parallel \text{canon}(E_n)). \quad (15.3)$$

The domain tags prevent cross-protocol ambiguity. A checkpoint may be digitally signed:

$$\sigma_n = \text{Sign}_{sk}(h_n \parallel g_n \parallel T_n). \quad (15.4)$$

A trusted timestamp authority or secure clock may be used when legal or evidentiary timing matters.



Changing an earlier canonical record changes every later chain head.

Figure 15.1: Tamper-evident forward timeline. Hash chaining detects modification when the chain head or signature is trusted.

15.3 Attachments and physical observations

A photograph, handwritten tag, location observation, or sensor snapshot becomes evidentiary only when the capture process, file bytes, timestamp, device identity, custody, and signature are controlled. The engine may store its attachment digest and context, but it may not claim that a visible sign by itself is a cryptographic proof or that a generated caption authenticates a location.

15.4 Identity data

The author declaration in this document is publication metadata. A deployed engine uses a protected private key, hardware-backed credential, or organizational certificate. It stores a pseudonymous public-key identifier where possible and separates identity from operational data.

Identity and cryptography

A name, birth date, civil identifier, vehicle number, officer number, or precinct label **MUST** not be used directly as a secret key or entropy source. A system that needs identity binding **MUST** use an authenticated credential and a documented key lifecycle, and **SHOULD** minimize exposure of raw personal identifiers.

15.5 What the provenance profile proves

A valid chain can show that a sequence of canonical records has not changed relative to a trusted chain head and signature. It does not prove that a sensor was truthful, a human interpretation was correct, a location was genuine, or an algorithm was fair. Those claims require independent controls and evidence.

Public-safety and responsible autonomy profile

16.1 Purpose

The author’s dedication motivates a public-safety profile for police and emergency-service contexts. The profile is limited to decision support, collision avoidance, bounded sensor processing, evidence integrity, emergency-signal localization, and accountable control. It does not grant autonomous coercive authority.

16.2 Candidate applications

Table 16.1: Public-safety application mappings.

Application	NHDF/PCKF role	Required evidence
Vehicle collision avoidance	Log-polar field, frame compensation, adaptive step, CCD, safe stop.	Missed-contact rate, false alarms, stopping envelope, sensor degradation tests, conventional baseline.
Emergency acoustic localization	Log magnitude, array phase, wavefront delay, attention interrupt.	Direction error, confidence calibration, multipath/noise tests, unknown-source-frequency handling.
Dash/body-camera quality	Exposure shock, ROI, frame stabilization, quality flags.	Raw/reconstructed image metrics, dropped coverage, audit logs, no unsupported anti-blooming claim.
Evidence provenance	Event digests, signatures, generation tags, attachment hashes.	Key security, canonicalization tests, clock behavior, custody procedures, independent verification.
Bounded sensor fusion	Shared cell state and quality flags across modalities.	Calibration, uncertainty propagation, failure-mode coverage, human-readable explanation.
Operator training/simulation	Replayable trajectories, branch/contact modes, telemetry.	Scenario validity, instructor oversight, no representation as operational certification.

16.3 Minimum-intervention control: “digital jiu-jitsu”

The commute transcript uses martial-arts language to describe redirecting momentum rather than meeting it with brute force. The formal counterpart is a minimum-intervention safety filter. Given a nominal action u_{nom} , choose

$$u^* = \arg \min_u \frac{1}{2} \|u - u_{\text{nom}}\|_R^2 \quad (16.1)$$

subject to actuator limits and safety constraints such as a control barrier condition

$$\dot{h}_j(x, u) + \alpha_j h_j(x) \geq 0, \quad (16.2)$$

where $h_j(x) \geq 0$ defines a safe set. If the problem is infeasible, the agent enters a documented safe state, ordinarily stop, retreat, or handover. This captures low-waste redirection while retaining explicit physics and authority limits.

16.4 Non-negotiable governance constraints

Public-safety constraint

A public-safety NHDF profile **MUST** satisfy all of the following:

1. A named human or legally accountable organization remains in command.
2. The engine does not independently authorize arrest, detention, use of force, pursuit escalation, or other coercive action.
3. Uncertainty, sensor quality, and failure state are visible to operators.
4. Data collection has a lawful purpose, documented retention, access control, and minimization policy.
5. Performance is tested across relevant populations, environments, lighting, acoustics, and mobility conditions; known disparities are reported.
6. Every safety-critical decision path is auditable and replayable within legal limits.
7. Cybersecurity includes signed software, key protection, rollback control, update provenance, and incident response.
8. A fail-safe mode exists for sensor conflict, deadline miss, resource exhaustion, integrity alarm, or model uncertainty.

16.5 Attention and surveillance boundary

The budgeted attention operator may prioritize safety-relevant regions, but it **MUST** not silently become unrestricted person tracking. Public-safety deployments declare the task, target class, retention period, access rights, and prohibited inferences. Where less intrusive sensing meets the safety objective, it is preferred.

16.6 No endorsement claim

The dedication and application profile do not state that Dutch police, the Hoofddorp police station, Roads Technology, ProBiblio, FLX Space, or any other named place or organization reviewed or endorsed this work.

Reference software and GPU realization

17.1 Semantics-first implementation order

The implementation priority remains:

1. deterministic CPU reference and replay vectors;
2. local field, log-polar Jacobian, phase, and contact correctness;
3. bounded allocation, generation tags, and failure telemetry;
4. GPU equivalence within tolerance;
5. profiling and selective optimization;
6. application proof against a baseline; and only then
7. vendor-specific or physical-substrate specialization.

17.2 Structure-of-arrays state

A practical GPU runtime separates:

- payload/parity/orientation and quality words;
- log-polar addresses and quantizer metadata;
- Cartesian apex, velocity, acceleration, and frame state;
- phase history and lock state;
- local-field parameters or references;
- parent, left, right, generation, free-list, and proposal queues;
- attention weights and mode state;
- contact gaps, normals, and response parameters;
- output/residual buffers; and
- event-record and digest staging buffers.

A dense address table may point to a compact active-cell pool. Contact, branch proposal, and provenance staging are explicitly bounded.

17.3 Reference kernel sequence

1. condition input: calibration, exposure, observer frame, quality;
2. compute causal residuals and log-polar addresses;
3. select budgeted active cells and apply declared transforms;
4. project local fields and compute parity/topology diagnostics;
5. update phase lock, debounce, modes, and branch proposals;

6. compact accepted proposals into the bounded pool;
7. integrate Cartesian kinematics and re-encode with J_{LP} checks;
8. perform conservative contact advancement and response;
9. evaluate sweeps/wavefronts and projection;
10. compute application residual and update state;
11. canonicalize telemetry and update the provenance chain.

17.4 Reference pseudocode

Listing 17.1: Semantics-first Version 0.2 loop.

```

for generation n in forward_timeline:
    conditioned, quality = condition_input(input[n], history, frame_state)
    address = log_polar_encode(conditioned, history)
    active = allocate_attention(address, quality, fixed_budget)

    parallel_for cell in active:
        z = apply_declared_transform(cell.candidate, cell.profile)
        z = boundary_project(z, cell.local_field)
        parity = parity_diagnostics(z.payload, z.chart_path)
        control = phase_lock_and_debounce(z, parity, quality)
        proposal[cell] = bounded_branch_proposal(z, control)

    accepted = compact_with_capacity(proposal, node_pool)

    parallel_for cell in active_cells:
        apply_links(cell, accepted)
        dt = min(nominal_dt,
                 resolution_step_bound(cell),
                 contact_step_bound(cell),
                 deadline_step_bound())
        integrate_cartesian(cell, dt)
        contact_guard_and_response(cell, dt)
        sweep_state[cell] = build_declared_wavefront(cell)

    output = project(sweep_state, view_or_application_parameters)
    residual = application_residual(output, observation_or_target)
    update_state_and_generation(output, residual, quality, accepted)
    append_hash_chained_event(telemetry_snapshot())

```

17.5 Memory budget

Let M_V be measured usable memory after display, driver, runtime, and fragmentation reserve. A conforming layout satisfies

$$M_{LUT} + M_{cell} + M_{branch} + M_{timeline} + M_{contact} + M_{output} + M_{provenance} + M_{reserve} \leq M_V. \quad (17.1)$$

The source’s illustrative 12 GB split remains a traceability example only. Version 0.2 adds explicit contact and provenance regions and requires runtime measurement rather than assuming all nominal VRAM is available.

17.6 Bitwise and matrix hardware

Population count and XOR naturally accelerate parity. Shifts may implement powers of two and fixed-point scaling. Matrix units may accelerate batched transforms or learned adapters. None of these units automatically implements topology, Möbius inversion, CCD, proof, or correctness. Low-level instructions are selected only after their exposed semantics match the reference operation.

18

Optical and quantum operator-equivalence extension

18.1 Physical transfer criterion

For an optical device with input field E_{in} and output E_{out} , an operator-equivalent stage satisfies

$$E_{\text{out}} = \mathcal{T}_{\text{phys}}(E_{\text{in}}) \approx \mathcal{O}(E_{\text{in}}) \quad (18.1)$$

within measured error, where $\mathcal{O}$ is a declared subset of the software chain. Phase, amplitude, polarization, spatial mode, and propagation distance may encode state, but the experiment must specify calibration, bandwidth, drift, energy, and repeatability.

18.2 Quantum criterion

A quantum state may encode an address/state superposition

$$|\Psi\rangle = \sum_i c_i |i\rangle |z_i\rangle. \quad (18.2)$$

A unitary/channel/measurement sequence must reproduce a declared operator and readout distribution within error. A classical local-zero-set constraint is not automatically a quantum error-correcting code, and the use of “holographic” does not imply quantum advantage.

18.3 Minimum physical experiment

A disciplined first experiment implements only one or two stages:

1. encode a known residual into optical or quantum coordinates;
2. apply a programmable transform or zero-set projection surrogate;
3. compare the measured output with the numerical reference;
4. report transfer error, drift, bandwidth, latency, energy, calibration, and failure rate.

Only measured operator equivalence justifies moving feedback, branching, or contact into the substrate.

19

Application adapters

19.1 Synthetic-aperture radar

The SAR profile maps a complex echo or estimated phase-error vector to the common state. Residual magnitude and phase enter the log-polar address; local zero sets represent valid focus or phase-consistency states; the router partitions range, azimuth, subaperture, scale, or hypothesis space; frame state represents platform motion; the wavefront stage models antenna support; and projection produces a focused image or correction field.

A severe ionospheric or hardware disturbance may raise jitter, phase acceleration, quantizer saturation, parity rate, and residual. The formal response is bounded safe mode, stronger integrity checking, uncertainty reporting, and comparison with a conventional autofocus baseline. The profile makes no claim of geomagnetic-storm immunity.

Minimum SAR metrics include phase-error RMS, image entropy or sharpness, point-target resolution, peak and integrated sidelobe ratios, runtime, memory, power, dropped frames, saturation, parity behavior, and failure under even-numbered bit faults.

19.2 Rendering and spatial simulation

A rendering adapter may use local fields, analytic cone sweeps, log-polar foveation, and the CCD profile. It reports image error, missed intersections, contact error, temporal stability, memory, and frame-time distribution. “Infinite detail” is approximated by bounded recursion, finite precision, and finite samples.

19.3 Robotics and navigation

A robotics adapter uses observer-frame compensation, conservative contact, minimum-intervention control, and safe mode. It separates perception uncertainty from dynamics and preserves an externally verifiable stop/retreat behavior. Soft-robot or articulated-tree mappings are candidate models, not structural identities.

19.4 Audio and spatialization

An audio adapter may route frequency-band states through the branch graph and map wavefront direction to output channels. Jitter and phase may modulate timbre, but signal quality, stability, clipping, latency, and hearing-safety constraints remain explicit.

19.5 Edge AI adapter

An AI adapter may use log-polar indexing, sparse cell activation, branch routing, or implicit parameterization. It does not follow that dense model weights disappear or that a model too large for memory will run automatically. Compression ratio, perplexity/accuracy, latency, energy, memory, calibration, and baseline quantization must be measured.

Verification program and falsifiable hypotheses

20.1 Core hypotheses

Table 20.1: Version 0.2 falsifiable hypotheses.

ID	Hypothesis	Minimum test
H1	Log-polar addressing reduces weighted quantization error for a declared residual distribution relative to an equal-capacity Cartesian or linear-magnitude table.	Matched-capacity Monte Carlo with calibrated distributions, saturation accounting, and confidence intervals.
H2	The Jacobian-correct kinematic update has lower physical trajectory error than independent radial/angular shifts.	Reference trajectories spanning radius, angular speed, cross terms, and apex clamp; compare position/velocity/energy error.
H3	Local zero-set projection contracts residual without destabilizing the integrator.	Sweep λ_B , step size, field conditioning, and noise; report convergence region and singular failures.
H4	Parity/phase control improves recovery latency or unnecessary-correction rate relative to always-on and threshold-only controllers.	Identical inputs and budgets; include even-bit faults and phase-unlock events.
H5	Golden-ratio/phase routing improves an application metric at equal node count.	Compare with fixed, optimized, random, and flat routers using preregistered metrics.
H6	Budgeted attention reduces compute while respecting a declared peripheral safety floor.	Measure latency/energy and missed events across focus shifts, occlusions, and adversarial saliency.
H7	Conservative advancement prevents tunneling within its declared assumptions.	Swept thin features, moving/deforming boundaries, acceleration, quantization, and known ground-truth contact times.
H8	Collision response matches the declared restitution/friction model.	Momentum/energy residuals, resting contact, sliding, stacking, and corner cases.
H9	Photometric shock handling improves recovery over a baseline without hiding saturation.	Raw-sensor or HDR sequence, tunnel/daylight transitions, quality flags, exposure latency, task performance.
H10	Acoustic adapter localizes an emergency-like signal with calibrated uncertainty.	Microphone-array data with source motion, multipath, noise, unknown source frequency, and negative controls.
H11	Hash chaining and signatures detect record modification and preserve forward order under wrap.	Canonicalization corpus, mutation tests, clock rollback, key rotation, partial-log loss, independent verification.
H12	CPU and GPU implementations are equivalent within declared tolerance and resource bounds.	Deterministic replay over transforms, branch overflow, contact, wraparound, modes, and failure injection.
H13	A public-safety profile improves a safety metric without unacceptable privacy, disparity, or operator-overreliance effects.	Independent operational simulation with human factors, legal/privacy review, subgroup analysis, and fail-safe evaluation.
H14	An optical or quantum stage reproduces a declared operator with useful error/energy tradeoff.	Physical transfer-function measurement against the numerical reference.

20.2 Required ablations

At minimum, experiments remove or replace one component at a time:

- log-polar metric correction versus raw shifts;
- local zero-set projection versus unconstrained state;
- identity versus Möbius, spiral, or hyperbolic transform;
- parity-only, phase-only, combined, and stronger integrity schemes;
- golden-ratio versus alternative branch parameters;
- fixed versus adaptive time step;
- discrete collision check versus conservative advancement;
- foveated budget versus uniform and task-optimized sampling;
- open-loop versus self-referential feedback;
- unsigned log versus signed checkpoints; and
- public-safety assistance with and without mandatory quality/failure display.

If a simpler ablation performs equally well, the removed operator is not justified for that application.

20.3 Telemetry contract

Every conforming run records at least

$$\{T_n, g_n, u_n, N_n, M_n, \Delta t_n, \max r_B, \|r_n\|, \text{parity rate, phase error, saturation, transform flags, attention budget, contact gap, CCD iterations, overflow, deadline miss, mode, runtime, energy if available, } h_n\}. \quad (20.1)$$

Application profiles add domain metrics and privacy/safety audit events.

20.4 Reference test vectors

The test suite **SHOULD** include:

- w at minimum, zero, and maximum; θ at wrap boundaries;
- pure radial, pure angular, and coupled motion;
- transform identity, involution, pole-near, and inverse tests;
- odd and even payload faults; odd and even topology reversals;
- phase ramps, chirps, wraps, and noisy second differences;
- branch capacity exactly full and one-over-capacity;
- contact grazing, head-on, sliding, high acceleration, thin barriers, and moving surfaces;
- dark-to-bright, bright-to-dark, clipped, and missing-frame sequences;
- acoustic single source, multiple sources, echoes, and source-frequency uncertainty;
- provenance mutation, reorder, truncation, clock rollback, and key rotation; and
- safe-mode entry, persistence, handover, and recovery.

Limitations and failure modes

1. **Degenerate field risk.** A literal global $F \equiv 0$ supplies no usable normal or distance.
2. **Non-orientable sign ambiguity.** Intrinsic charts or unsigned/local constraints are required; a global inside/outside label may not exist.
3. **Log-polar singularity.** The apex requires a finite clamp, sentinel, or alternate chart.
4. **Metric omission.** Raw address shifts can create radius-dependent physical error and omit centripetal/cross terms.
5. **Transform poles and encoding mismatch.** Möbius or hyperbolic operators can diverge; bit instructions may not implement their mathematics.
6. **Parity blind spots.** One-bit parity misses even fault counts and corrects nothing.
7. **Phase noise.** $\Delta^2\phi$ amplifies noise and wrap errors.
8. **Hybrid chattering.** Parity, phase, attention, shock, and contact modes require hysteresis and rate limits.
9. **Branch explosion.** Recursive rules require hard capacity, compaction, and overflow policy.
10. **CCD assumptions.** Invalid distance/rate bounds, deforming geometry, discontinuities, or finite precision can cause missed contact.
11. **Collision-model mismatch.** Restitution and friction parameters may be uncertain; fragmentation can violate conservation or memory bounds.
12. **Sensor physics.** Coordinate transforms cannot recover clipped values, prevent physical blooming, or remove all rolling-shutter artifacts.
13. **Attention blind spots.** Foveation can miss peripheral events and can encode bias if saliency is poorly chosen.
14. **Acoustic ambiguity.** Multipath, unknown source signal, and synchronization limit Doppler and direction estimates.
15. **Provenance scope.** A valid hash chain proves consistency relative to a trusted head, not truth of the captured world.
16. **Identity/privacy risk.** Raw personal data can become a security and legal liability; it is not entropy or proof.
17. **Topology is not resilience.** A Klein-bottle quotient does not protect hardware from radiation, memory faults, malware, or corrupted inputs.
18. **Hardware mismatch.** Tensor, ray, optical, or quantum units help only when exposed operations match the formal operator.
19. **Application dependence.** Projection shapes, focus objectives, safety metrics, and ethical constraints are adapters, not universal outputs.
20. **Institutional use.** Public-safety adoption requires legal authority, procurement validation, cybersecurity, training, human factors, and independent oversight beyond this document.

Normative minimum specification, Version 0.2

An implementation may call itself an NHDF/PCKF Version 0.2 engine only when it satisfies all **MUST** requirements below.

1. It **MUST** define a monotonic logical timeline, generation tags, elapsed-time semantics, and ring-buffer rules.
2. It **MUST** define causal input residuals, log-polar encoding, quantization, saturation, inverse interpretation, and quality flags.
3. It **MUST** distinguish jitter log magnitude ρ_J from spatial log radius w and phase from polar angle.
4. It **MUST** implement or explicitly bypass the log-polar Jacobian and state how address-space rates become physical motion.
5. It **MUST** regularize the apex singularity with a finite clamp, sentinel, or chart.
6. It **MUST** represent “SDF zero” as non-degenerate local constraints or document an equivalent formulation.
7. It **MUST** define parity inputs, topology orientation semantics, blind spots, debounce, and safe-mode behavior.
8. It **MUST** define phase unwrapping/filtering, $\Delta^2\phi$, and phase-lock behavior when those operators are used.
9. It **MUST** bifurcate only at declared branch nodes, preserve declared BST ordering, and bound all node allocation.
10. It **MUST** declare exactly how φ_g and $\Delta^2\phi$ affect branch geometry or priority.
11. It **MUST** integrate forward kinematics with a declared numerical method and time-step bound.
12. A CCD profile **MUST** declare field validity, rate bounds, tolerances, contact localization, response law, iteration limit, and failure fallback.
13. Every optional coordinate transform **MUST** declare its equations, domain, singularities, orientation effect, and register-equivalence test.
14. An attention profile **MUST** declare its budget, focus selection, peripheral floor, dropped coverage, and recovery path.
15. A photometric or acoustic profile **MUST** expose saturation, uncertainty, and sensor-model assumptions.
16. It **MUST** define analytic sweep/wavefront and projection operators and the application residual.
17. It **MUST** feed output or residual into the next state generation.
18. It **MUST** bound cells, branches, histories, contact work, output workspace, and runtime reserve.
19. It **MUST** emit telemetry and a visible failure state when bounds or assumptions are violated.
20. A provenance profile **MUST** use canonical records, a cryptographic hash, protected keys for signatures, and privacy-preserving identity references; raw identity is not key material.
21. A public-safety profile **MUST** comply with the human-authority, no-autonomous-force, privacy, audit, fairness, cybersecurity, and fail-safe requirements in chapter 16.
22. It **SHOULD** provide deterministic replay, reference vectors, CPU/GPU equivalence tests, and independent validation.
23. It **SHOULD** keep speculative optical, quantum, SAR, AI, robustness, and institutional claims separate from core conformance.

Compact formal definition

A Version 0.2 NHDF engine is a bounded causal hybrid state machine

$$\mathfrak{N}_{\Delta t} : X \times L \times H \rightarrow L \times H$$

whose transition is the closed composition

$$\begin{aligned} y_n &= \Pi \circ \mathcal{S} \circ \mathcal{C}_{\text{CCD}} \circ \mathcal{K}_{T \rightarrow}^J \circ \mathcal{R}_{\text{BST}}^{\varphi_g, \Delta^2 \phi_n} \circ \mathcal{C}_\phi \circ \mathcal{P} \circ \mathcal{B}_0 \\ &\quad \circ \mathcal{X}_{\xi_n} \circ \mathcal{A}_B \circ \mathcal{E}_{\text{LP}} \circ \mathcal{C}_{\text{in}}(x_n; L_n), \\ L_{n+1} &= \mathcal{U}(L_n, y_n, r_n), \\ h_{n+1} &= \mathcal{H}(h_n, E_{n+1}), \end{aligned}$$

with cell-local validity, finite resources, monotonic logical time, explicit metric and parity semantics, measurable residual, exposed failure, and profile-specific constraints.

Conclusion

Version 0.1 made the original prompt chain coherent by treating it as a typed operator specification. Version 0.2 extends that work with the useful engineering knowledge developed during the commute to Hoofddorp.

The most important addition is the conformal log-polar metric. A radial or angular bit shift is an address operation; physical velocity and acceleration require radius scaling, a Jacobian, centripetal terms, and radial-angular coupling. Once this is explicit, Möbius inversion, spiral shear, observer-frame compensation, and optional hyperbolic transport can be treated as real coordinate maps rather than names attached to gates.

The second major addition is bounded contact. Local fields can support conservative advancement and collision response, but only under declared distance, gradient, time-rate, integration, and precision assumptions. Restitution, friction, sliding, and fragmentation are physical/application policies, not consequences of parity.

The third addition is operational accountability. Attention is a budget allocation with blind-spot controls; exposure and acoustics are sensor adapters with uncertainty; photographs and signs become attachment digests rather than self-authenticating proof; and public-safety use is constrained by human command, privacy, auditability, fairness, cybersecurity, and fail-safe operation.

The upgraded paradigm is therefore a sharper scientific proposal: a finite, causal, closed operator machine that can be implemented, replayed, ablated, and disproved. Its value will be established not by the breadth of its metaphors or the identity of its author, but by reproducible advantages attributable to its ordered composition and by responsible behavior when its assumptions fail.

A

Expanded symbol table

Symbol	Meaning
$x_n, \bar{x}_n, e_n, J_n$	Input, causal baseline, residual, and residual magnitude.
ρ_J, θ, T	Log jitter magnitude, selected polar direction, and forward time address.
w, r, r_0, b	Log spatial radius, physical radius, reference radius, and log base.
J_{LP}, G_{LP}	Log-polar Jacobian and induced metric.
L_n, c_i, q_i	Complete state, cell record, and packed payload.
a_i, v_i, α_i	Distributed apex position, velocity, and acceleration.
φ_g	Golden ratio scalar.
$\phi_i, \Delta^2 \phi_i$	Signal/cell phase and second finite difference.
$F_i, M_i, r_{B,i}$	Local implicit constraint, its zero set, and boundary residual.
$p_i^{\text{data}}, p_i^{\text{topo}}, \bar{p}_i$	Payload parity, orientation parity, and debounced gate.
o_i	Local chart orientation sign.
$\kappa_i, N_{\max}$	BST ordering key and node capacity.
$\mathcal{C}_{\text{in}}$	Calibration, photometric, and observer-frame conditioning.
$\mathcal{E}_{LP}$	Log-polar residual/address operator.
$\mathcal{A}_B$	Budgeted attention/resource allocator.
$\mathcal{M}_{c,R}$	Möbius inversion profile.
Γ_κ	Log-spiral shear profile.
$\mathcal{G}_c$	Optional hyperbolic gyrotranslation.
$\mathcal{B}_0$	Local zero-set projection.
$\mathcal{P}$	Parity/topology reduction.
$\mathcal{C}_\phi$	Phase lock, debounce, orientation, and safe-mode controller.
$\mathcal{R}_{\text{BST}}$	BST/L-system branch router.
$\mathcal{K}_{T \rightarrow}^J$	Forward physical kinematics with log-polar metric.
$\mathcal{C}_{\text{CCD}}$	Conservative contact and collision-response operator.
$\mathcal{S}, \Pi$	Wavefront/implicit sweep and observation projection.
$\mathcal{U}$	Self-referential state update.
E_n, h_n, σ_n	Canonical event record, chain head, and optional signature.
$\mathcal{H}$	Provenance-chain update.
$\mathfrak{N}_{\Delta t}$	Complete Version 0.2 transition.

B

Proof and property sketches

B.1 Non-degeneracy

If one global differentiable F is identically zero, every directional derivative is zero. No normal or restoring direction can be obtained. Therefore any normal-based implementation needs nontrivial local fields, constraint Jacobians, or multiple charts.

B.2 First-order boundary contraction

Let $g = \nabla F(z)$ and $z^+ = z - \lambda F(z)g/(\|g\|^2 + \varepsilon)$. Taylor expansion gives

$$F(z^+) = F(z) \left(1 - \lambda \frac{\|g\|^2}{\|g\|^2 + \varepsilon} \right) + O(\|z^+ - z\|^2). \quad (\text{B.1})$$

For small residual and $0 < \lambda < 2$, the leading term contracts when the gradient is well conditioned.

B.3 Log-polar acceleration

With $r = r_0 b^w$, $\dot{r} = (\ln b)r\dot{w}$, and $\ddot{r} = r((\ln b)\ddot{w} + (\ln b)^2\dot{w}^2)$. Substituting into the standard polar acceleration

$$a = (\ddot{r} - r\dot{\theta}^2)e_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})e_\theta \quad (\text{B.2})$$

produces the law in chapter 5. This proves that scale and cross terms cannot be represented by angle alone.

B.4 Möbius inversion in log-polar coordinates

For $z = re^{i\theta}$, $1/z = r^{-1}e^{-i\theta}$. Taking $w = \log_b(r/r_0)$ yields $w' = -w$ up to the declared reference-scale constant and $\theta' = -\theta$. Applying the map twice returns the original state away from the pole.

B.5 Bounded memory

If every cell and branch allocation is drawn from finite pools, every proposal is accepted only when capacity exists, and rejected/recycled proposals create no hidden allocation, then $N_n \leq N_{\max}$ by induction.

B.6 Parity detection

XOR parity is addition modulo two. It changes under an odd number of bit inversions and remains unchanged under an even number; therefore it is an incomplete integrity mechanism.

B.7 Causal ring buffer

If each physical slot stores generation g , a read for generation g_n rejects any slot tagged above g_n . Wraparound reuses storage without violating logical order.

B.8 Conservative contact step

If the gap rate satisfies $|\dot{F}| \leq B_F$ and $F(t) = g > 0$, then over $\Delta t \leq \eta g / B_F$ the worst-case gap reduction is at most $\eta g < g$, so contact cannot be crossed in that step under the stated bound. Discretization and model uncertainty motivate the tolerance and safety factor.

B.9 Hash-chain tamper evidence

If H is collision resistant and canonical records are unambiguous, changing any earlier event changes its digest input and therefore every later chain head except with negligible collision probability. Trust still depends on the protected chain head/signature and truthful capture.

C

Traceability to the supplied sources

Source location	Development	Version 0.2 destination
[S0], pp. 1–6	Original vocabulary, kinematics, parity-conditioned boundary restoration.	Chapters 2–4, 6, 9, and 13.
[S0], pp. 8–18	Operator precedence, SDF moved into LUT, distributed apex, NHD-F/IHCP naming.	Chapters 1–4 and 15.
[S0], pp. 19–30	$\Delta^2\phi$, forward timeline, optical/quantum, SAR, GPU.	Chapters 6, 9, 18, 19, and 20.
[S1], all	Baseline Version 0.1 formal model and evidence boundary.	Retained and revised throughout Version 0.2.
[S2], pp. 1–4 (07:50–07:59)	Time edge, missing radial scale/-Jacobian, cross-coupling.	Chapters 2, 5, and 8.
[S2], pp. 5–8 (08:03–08:13)	Möbius inversion, gyrovectord shift, phase lock, frame drift, ray twist, temporal guard.	Chapters 6, 7, 8, and 10.
[S2], pp. 8–10 (08:16)	Focal attenuation, interrupt, coherence recovery.	Chapter 9.
[S2], pp. 10–17 (08:19–08:25)	Register logic, CCD interpretation, sliding, damping, fragmentation.	Chapter 12 and GPU chapter 17.
[S2], pp. 17–21 (08:29–08:34)	Foveal resource concentration, dark/bright transition, narrow visual mask.	Chapters 7 and 9.
[S2], pp. 21–23 (08:37)	Police siren, Doppler, acoustic interrupt.	Chapter 13 and public-safety profile.
[S2], pp. 23–25 (08:40–08:42)	Soft/diffuse fields, stationary origin, acceleration/deceleration claim.	Chapters 5, 7, and correction box in Chapter 5.
[S2], pp. 26–30 (08:44–08:50)	Blackboard metadata, identity salt, one-dimensional timeline.	Chapter 15, with cryptographic and privacy corrections.
[S2], pp. 31–36 (08:52–08:57)	Edge applications, police use cases, digital jiu-jitsu, autonomous agents.	Chapters 16 and 20, with human-authority and no-autonomous-force constraints.
[S2], pp. 37–40 (09:00–09:03)	Anisotropic mask, final state commit, stationary work mode.	Chapters 7, 10, and 15.

D

Source notes and version history

Source notes

- [S0] 30-page exported prompt-and-response transcript beginning with “1bit parity bit lower case phi jitter log encoded polar LUT...”, supplied by the author.
- [S1] *Non-Euclidean Holographic Data Fields: A Formal Operator Specification for a Parity-Conditioned Kinematic Foliation and Implicit Holographic Co-Processor*, Version 0.1, Tom Klootwijk, 31 August 2026.
- [S2] 40-page exported commute transcript beginning at approximately 07:50 with boarding the tram at IJburg and ending at approximately 09:03 at the Hoofddorp work location, supplied by the author.

Version history

Version	Date	Summary
0.1	31 Aug 2026	First non-degenerate formalization of NHDF, PCKF, and IHCP; local zero sets, typed chain, bounded routing, GPU/optical/SAR extensions, validation program.
0.2	31 Aug 2026	Roads Technology commute extension: conformal log-polar metric, transform profiles, observer/photometric conditioning, attention/interrupt, adaptive timeline, CCD/response, acoustics, provenance, responsible public-safety profile, expanded tests and corrections.